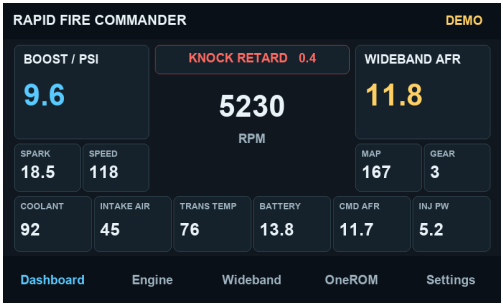
RAPID FIRE SOFTWARE

Rapid Fire Commander

Wired handbook

Windows flashing, OneROM, TunerPro RT, SD tunes and wideband wiring.



Instrument Cluster, illustrated from the firmware layout

RapidFire-001 / rapid-fire-commander

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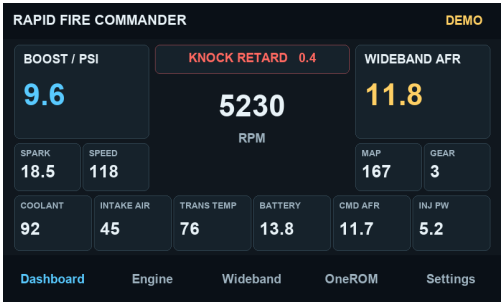
Print as a bifold booklet. In a PDF reader with booklet printing, choose **Booklet**, A4 paper, both sides and left binding. If the printer requires a flip setting, choose short-edge duplex. Fold the sheets together down the middle. The pages in this PDF are in reading order, so booklet printing handles the sheet order.

Overview and support

Rapid Fire Commander brings an instrument cluster, OneROM tune selection and TunerPro RT connectivity to the Guition 4.3-inch ESP32-P4 display.

Edition	Windows connection	Printable A5 guide
Wired	USB3 serial bridge	Wired booklet
Wi-Fi	Commander Wi-Fi and the Windows Companion launcher	Wi-Fi booklet

Choose the kit for your hardware on the Releases page. Check the asset names for Wired or WiFi. Older releases may only provide Wired.



Instrument Cluster in Demo mode

The Dashboard groups RPM, boost, wideband AFR and knock retard around the centre, with supporting readings for MAP, gear, speed, spark, temperatures, battery, commanded AFR and injector pulse width. **Settings > Demo mode** animates the gauges and marks the screen **DEMO**. Live mode is the startup default. A dash means no valid live value is available.

The **Engine** tab offers **Auto**, **11P** and **12P**. Auto follows a supported tune family after a verified SD tune load. Choose the matching ADX variant and MAP range in TunerPro RT for ECU logging.

- Install or update on Windows with Espressif Flash Download Tool
- Dashboard, Engine, OneROM, SD card and TunerPro help
- Wideband wiring and source selection

- Firmware licence

Print the PDF for your edition using Booklet mode on A4. For support, email Rapid Fire Software Support with your edition, version, board model and the exact error message.

Print a folded copy

Print the whole PDF in Booklet mode on A4. Use both sides and left binding, then fold the sheets together. Keep this handbook with the Commander so the wiring and recovery steps are easy to find.

Flash on Windows

Choose the package and check the board

Use the Wired or Wi-Fi package for your device. Both target the Guition JC4880P443C_I_W/Y 4.3-inch ESP32-P4 display with 16 MB flash and P4 silicon revisions v0.1 through v1.99. Verify the board label and revision before flashing. Disconnect OneROM and vehicle wiring.

The merged BIN replaces the P4 flash, including saved settings. Back up the existing firmware before updating if you need a recovery image. The Wi-Fi package updates the P4 and relies on the existing wireless coprocessor firmware.

Flash with Espressif Flash Download Tool

1. Extract the entire release ZIP. Verify the merged BIN against the supplied SHA-256 checksum with PowerShell's **Get-FileHash**.

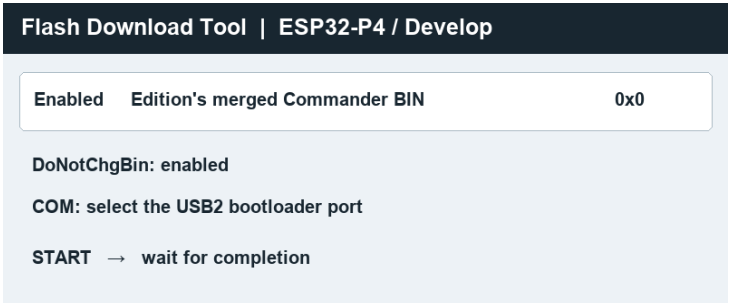
2. Open Windows Device Manager and expand **Ports (COM & LPT)**. With the board unpowered, hold **BOOT**, connect a data-capable cable to **USB2**, and release BOOT when the new COM port appears.

3. Open **Flash Download Tool**. If it is absent from your kit, download it from Espressif. Choose **ESP32-P4** and **Develop**.

4. Select the edition's merged BIN on a single enabled row at **0x0**. Clear other rows and enable **DoNotChgBin**. Select the USB2 bootloader COM port.

5. Click **START** and keep the USB cable connected until the tool reports completion. Reconnect power with BOOT released to start Commander.

6. Check the Dashboard and Engine tab. Turn on **Settings > Demo mode** to check gauge formatting and turn it off again for live use.



Illustrated Flash Download Tool settings

See the ESP32-P4 Flash Download Tool guide for the tool screens.

Connect after flashing

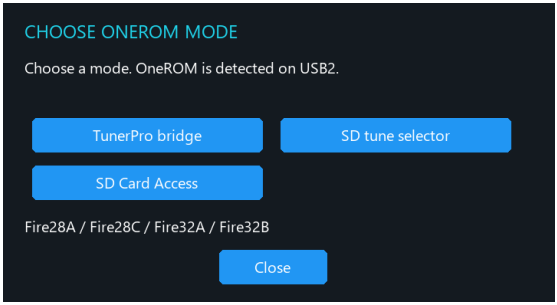
Edition	Connection
Wired	Connect Windows to **USB3** for the TunerPro bridge or SD Card Access. In normal operation **USB2** hosts OneROM.
Wi-Fi	Connect Windows to the Commander network and start the bundled Windows Companion launcher. Map the local RFC 2217 endpoints to COM ports with your Windows virtual serial port provider. Use its OneROM mapping at **127.0.0.1:22170** and the separate GT Pro mapping at **127.0.0.1:22171** when enabled. Use USB2 for P4 flashing.

Troubleshooting

Symptom	Check
No new bootloader COM port	Try a known data cable directly to Windows, check USB2 and repeat the BOOT sequence.
COM port busy or unable to connect	Close TunerPro and serial monitors. Recheck the bootloader port in Device Manager.
Flash fails	Verify the edition and checksum, check power and the cable, then retry in bootloader mode.
Blank screen after flashing	Boot with BOOT released. Verify the board model and flash size before retrying.
No live ECU readings	Verify the matching ADX in TunerPro RT and check the acquisition connection. The Engine family selector identifies a profile, while TunerPro decodes the ECU data.

Commander menus and tuning

The illustrations are recreated from the firmware layout. Status values and example filenames illustrate the workflow.



OneROM mode chooser with TunerPro Bridge, SD Tune Selector and SD Card Access

Program and verify OneROM in the Rapid Fire OneROM app

OneROM firmware mode	Use
NORMAL DT	Honda standard Ostrich with Display Trace. The optional UART runs at 38,400 baud.
ALDL 808 DT	Delco 808 wired directly to OneROM for ALDL at 8,192 baud, with default signal polarity.

1. Keep OneROM disconnected from the ECU. Connect it directly to your computer and open the Rapid Fire OneROM app.

2. The Rapid Fire OneROM app automatically detects and selects your OneROM board and sets the intended ROM size. For a Delco 808 wired directly to OneROM, select **ALDL 808 DT** in the app. Then program your starting tune BIN. If you log through GT Pro, use its separate acquisition COM port and the matching ADX in TunerPro RT.

3. Read the programmed ROM back in the app and confirm that it matches the intended BIN byte for byte. If you plan to use the SD Tune Selector, put a copy of that verified BIN at the root of the FAT32 microSD card.

4. Once the readback matches, disconnect OneROM from the computer and plug it into Commander's **USB2** port. Open the Commander mode you want to use and confirm that it detects the expected board and ROM size.

Choose a Windows connection

Wired uses USB3. Wi-Fi uses the Commander network and the Windows Companion launcher. Its local endpoints are 127.0.0.1:22170 for OneROM and 127.0.0.1:22171 for GT Pro when enabled. The USB3 instructions below describe Wired. See the Windows guide before opening TunerPro.

OneROM modes

Mode	Use
TunerPro bridge	Laptop on USB3, OneROM on USB2. Select the P4's Ostrich COM port in TunerPro RT for emulation.
TunerPro + GT Pro	Separate Windows COM ports for OneROM emulation and read-only GT Pro ADX logging.
SD tune selector	Browse and load a matching BIN from the FAT32 microSD card onto OneROM with the engine off.
SD Card Access	Expose the P4's SD card as a removable drive through USB3. The tune selector and serial bridge pause while this mode owns the card.

TunerPro bridge

Connect the laptop to USB3 and the OneROM to USB2. Open **OneROM > TunerPro bridge**. Wait for the board and size to be detected, then find the new COM port in Windows and select it in TunerPro RT for Ostrich emulation. Start with a readback before attempting a write. Some OneROM profiles expose an optional second UART port. For GT Pro telemetry, choose **TunerPro + GT Pro**.

With a compatible OneROM setup and emulation enabled, TunerPro RT can send live calibration edits while the engine is running. TunerPro RT sends the editor update through Commander's bridge, and OneROM serves the updated data to the ECU. Perform live edits while the vehicle is stationary. See the TunerPro RT emulation guide.



TunerPro Bridge waiting for OneROM on USB2

TunerPro + GT Pro ADX logging

Commander passes read-only ALDL requests and replies between TunerPro RT and the GT Pro. Load the ADX in TunerPro RT so it can decode the ECU replies.

1. Connect OneROM to **USB2** and the Windows laptop to **USB3**. On Commander, choose **OneROM > TunerPro + GT Pro**.

2. In **Settings > OBDX devices**, select the GT Pro and tap **Connect selected**. Wait for **GT Pro ALDL acquisition ready** on the OneROM screen.

3. In TunerPro RT, use the **OneROM emulation** COM port for Ostrich. Under **Preferences > Data Acq./Emulation**, choose **Standard Serial** and the separate **GT Pro ALDL acquisition** COM port. Windows assigns the port numbers.

4. Stop acquisition if it is running. Choose **Acquisition > Load Definition File**, load the ADX for your ECU and calibration, then start acquisition.

After a verified SD tune load, Commander can identify OSE $11P V104 or $12P V112. Use that recommendation to choose the matching ADX variant in TunerPro RT. For a factory or unknown profile, choose the definition for the installed ECU and calibration.

Software emulation passed 35 monitor polls across 27 Holden ADX definitions, including OSE $11P, OSE $12P, VS Enhanced and the stock VS V6 $51 profile. The owner also reported validating GT Pro Bluetooth and Windows COM port detection on the physical P4. See the Commander v0.13 release notes.

SD Tune Selector

The SD Tune Selector loads the selected BIN image in blocks. Even when two BINs share the same ECU program code, an ECU reading during the transfer can see a mix of old and new calibration data. For SD tune swaps and full BIN uploads, OneROM can stay installed in the vehicle's MemCal. Keep the engine off throughout the upload and verify the readback before starting it.

Fire model	Supported ROM size
Fire28A, Fire28C	16, 32 or 64 KiB
Fire32A, Fire32B	128 KiB, 27C010

Commander uses model and active-size metadata from OneROM. Unsupported or unreadable profiles stop before a load. BIN transfers have been verified with **NORMAL DT**. Commander operation with **ALDL 808 DT** and end-to-end Display Trace in TunerPro RT still need verification.

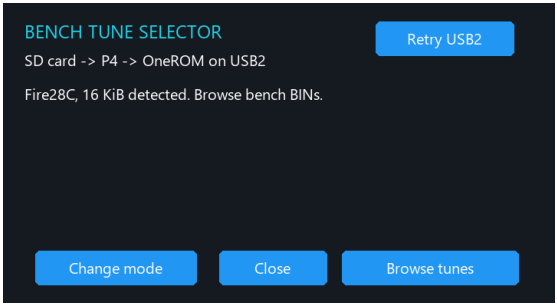
1. Copy the candidate BINs and a verified copy of the current BIN to the root of a FAT32 card. Use descriptive filenames.

2. Choose **SD tune selector** and check the detected board and ROM size.

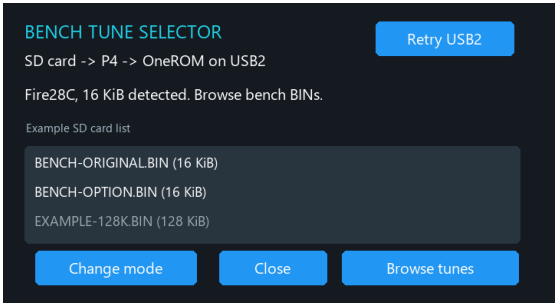
3. Tap **Browse tunes**. The list shows up to 64 BIN filenames and sizes. Files of a different size are disabled.

4. Select a compatible file and confirm the load. Commander checks the existing image against the card and validates the expected ROM layout before writing. For 32, 64 and 128 KiB images, identical halves fail this check.

5. Wait for **Verified byte for byte**. If a write is incomplete or verification fails, keep the engine off and restore and verify the original BIN before starting it using your OneROM recovery procedure.



Fire28C SD Tune Selector ready to browse

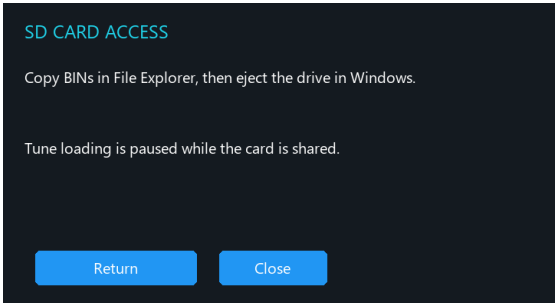


Example SD BIN list with a disabled size mismatch

Tap **Retry USB2** if the selector loses the device. The P4 can restart its USB2 host and the OneROM session while the cable stays in place. A retry does not replay a write. For **Change mode**, finish transfers, close TunerPro, turn the engine off and confirm the restart. A restart can discard unsaved tune changes.

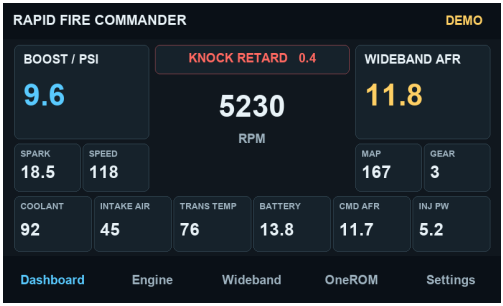
SD Card Access

Choose **SD Card Access** and connect Windows to USB3. Copy files to the card in File Explorer. Eject the drive in Windows and wait for the P4 to show that the card was released. Tap **Return** and select the next mode after the restart. You can also copy files with a separate card reader.



SD Card Access with Windows eject instructions

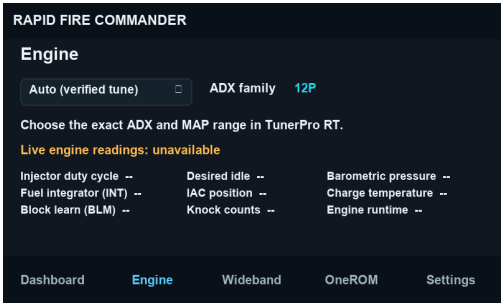
Dashboard, Engine and Settings



Instrument Cluster in Demo mode

• **Dashboard** shows RPM, boost, MAP, speed, gear, spark advance, knock retard, commanded AFR, wideband AFR, injector pulse width, coolant, intake and transmission temperatures, and battery voltage. Unavailable readings display dashes. A supported external wideband can feed the wideband display directly.

• **Engine > Auto** uses the family detected after a verified SD tune load. Choose **11P** or **12P** to review the corresponding channels manually. Select the exact ADX and MAP range in TunerPro RT. The Engine selector does not decode an ADX or supply missing live readings.



Engine family selector

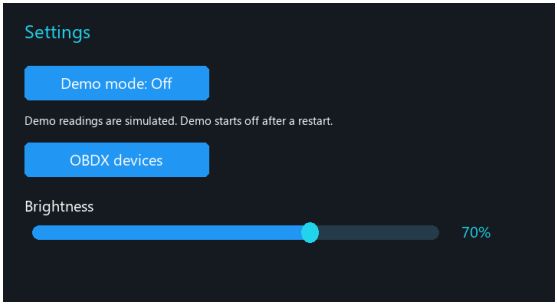
• **Settings > Brightness** adjusts the backlight from 10% to 100% and saves on release.

• **Settings > Demo mode** supplies simulated gauge values, marked by a **DEMO** badge. It starts off after a restart.

• **Settings > OBDX devices** opens Bluetooth discovery, connection controls and diagnostics. Use **TunerPro + GT Pro** to log ECU data in TunerPro RT.

• **Wideband > Source** chooses Innovate LM-2 or AEM 30-0300. See wideband setup for wiring and signal levels. Use one source at a time.

Settings and startup screens



Settings with Demo mode and brightness



Rapid Fire Commander startup logo fade

Wideband wiring

Commander receives a digital wideband signal on **P4 UART2 RX, GPIO32**. Use one supported source at a time. The screen shows AFR and lambda when valid data arrives.

Both editions use the same P4 serial input. Turn Demo mode off before checking an actual sensor reading.

Find the wideband input on the display board

On the Guition **JC4880P443C_I_W** schematic, **JP1** is the 2×13 header labeled **Expand IO**. Look for **GPIO32** at JP1 pin 19 and **GND** at pin 5. The schematic shows pin 20 beside pin 19 as **C6_U0RXD**. That pin belongs to the ESP32-C6. The separate RS-485 connector uses different signals. Match the board model and pin-1 orientation to the drawing before wiring. Guition Expand IO schematic

Required serial interface

Use a **true RS-232 to 3.3 V UART receiver**, for example a MAX3232 breakout designed for 3.3 V operation and powered at 3.3 V. Check the breakout pinout for its RS-232 receive input and its 3.3 V logic receive output. Keep raw RS-232 voltages, 5 V logic, vehicle power, CAN and analog outputs away from the P4 GPIO. A USB-to-TTL cable alone does not convert RS-232 signal levels. MAX3232 reference

Wideband controller wiring



Solid arrows show data. Dashed lines show ground. Leave RS-485 unused.

From	To
Wideband RS-232 TX	Receiver RS-232 input, often labeled R1IN

From	To
Receiver 3.3 V logic output, often labeled R1OUT	Guition JP1 pin 19, P4 GPIO32
Receiver logic ground	Guition JP1 pin 5, GND
Wideband serial signal ground	Receiver RS-232 signal ground, according to the controller and receiver pinouts

The P4 connection is receive-only. Leave adapter transmit wiring to the wideband unused. Power the module from a suitable regulated 3.3 V source as specified by the module. Check the JP1 orientation and actual board revision with power off. Guition board specifications

Choose the source

Controller	Signal	Commander setting
Innovate LM-2	MTS Serial OUT. Use the matching MTS serial-out lead for the LM-2 connector and verify its TX and ground pinout.	**Wideband > Source: LM-2**. The firmware listens at 19,200 baud, 8N1 and uses the first channel.
AEM X-Series 30-0300	Blue RS-232 Serial OUT wire. Set the gauge to AFR display mode.	**Wideband > Source: AEM**. The firmware listens at 9,600 baud, 8N1 and derives lambda using a gasoline reference of 14.65.

Compare the P4 reading to the controller display on the bench. Dashes indicate unavailable input or invalid readings. The dashboard returns to LM-2 after a reboot. This feature displays readings on the P4. TunerPro wideband passthrough remains pending, and both sources still need physical verification on this build. Innovate LM-2 manual · AEM 30-0300 manual

Firmware licence

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Follow the Windows flashing guide and back up the existing firmware. For OneROM writes, follow the bench procedure with the ECU disconnected.

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